

Land Survey

Engineering Mathematics III

Assignment 1 - Questions Only

Instructions: Answer all questions. Show formula, substitution, calculation, units and interpretation.

Section A: Formula and Concept Questions

1. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\text{Mean: } \bar{x} = \Sigma x/n$$

2. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$s = \sqrt{[\Sigma(x - \bar{x})^2/(n - 1)]}$$

3. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$P(A) = n(A)/n(S)$$

4. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$AX = B$$

5. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\text{error} = \text{observed} - \text{true}$$

Section B: Applied Questions

Problem Set 1

6. Solve a land survey problem involving repeated angle observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)
7. Solve a land survey problem involving distance measurement errors. Show formula, substitution, correct units and a technical interpretation. (10 marks)
8. Solve a land survey problem involving survey adjustment. Show formula, substitution, correct units and a technical interpretation. (10 marks)
9. Solve a land survey problem involving quality-control checks. Show formula, substitution, correct units and a technical interpretation. (10 marks)
10. Solve a land survey problem involving probability of acceptable observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 2

11. Solve a land survey problem involving repeated angle observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)
12. Solve a land survey problem involving distance measurement errors. Show formula, substitution, correct units and a technical interpretation. (10 marks)
13. Solve a land survey problem involving survey adjustment. Show formula, substitution, correct units and a technical interpretation. (10 marks)
14. Solve a land survey problem involving quality-control checks. Show formula, substitution, correct units and a technical interpretation. (10 marks)
15. Solve a land survey problem involving probability of acceptable observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 3

16. Solve a land survey problem involving repeated angle observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)
17. Solve a land survey problem involving distance measurement errors. Show formula, substitution, correct units and a technical interpretation. (10 marks)
18. Solve a land survey problem involving survey adjustment. Show formula, substitution, correct units and a technical interpretation. (10 marks)
19. Solve a land survey problem involving quality-control checks. Show formula, substitution, correct units and a technical interpretation. (10 marks)
20. Solve a land survey problem involving probability of acceptable observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 4

21. Solve a land survey problem involving repeated angle observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)
22. Solve a land survey problem involving distance measurement errors. Show formula, substitution, correct units and a technical interpretation. (10 marks)
23. Solve a land survey problem involving survey adjustment. Show formula, substitution, correct units and a technical interpretation. (10 marks)
24. Solve a land survey problem involving quality-control checks. Show formula, substitution, correct units and a technical interpretation. (10 marks)
25. Solve a land survey problem involving probability of acceptable observations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.
2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.
3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.
4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.